Gender-Associated Bias in Neural Rerankers — An Empirical Audit and De-Identification Test

Vasyl Rakivnenko
Stanford University
vasyl@stanford.edu

Abstract

Automated hiring pipelines can retrieve candidate documents with an embedding model and then reorder the leading candidates with a neural reranker. Bias in embedding models is well documented, but rerankers are a different class of model and have received little counterfactual auditing. We audit 14 open and proprietary rerankers with 8,320 counterfactual document pairs per model. Within each pair, qualifications are unchanged and only the selected name and pronouns differ. All 14 models favor the male-associated document more often for male-typed occupations than for female-typed occupations. In synthetic pools of 20 equally qualified candidates, one retrieval-and-reranking pipeline assigns 79% of top-3 positions to male-associated documents; behind the same retriever, the audited rerankers range from 44% to 85%, against 60% for retrieval alone. Results vary materially across models, occupations, names, templates, and query wordings. We then test a simple input transform that replaces names with one placeholder and gendered pronouns with one uniform token before ranking. The transform makes each counterfactual pair textually identical and therefore removes score differences attributable to those markers by construction. On a near-ceiling synthetic occupation-matching check, top-rank accuracy is 827 of 832 before and 830 of 832 after transformation, a difference not distinguishable from zero. These findings establish sensitivity to the tested names and pronouns, rather than discrimination in deployed hiring or a complete solution to demographic bias.

1 Introduction

Modern search systems commonly retrieve a candidate set with a fast first-stage method, such as lexical matching or a dense embedding model [5], and then use a more expensive model to rerank that set [6]. The same architecture can be used for job and resume matching. A candidate must therefore pass two learned ordering stages before appearing near the top of a recruiter's list.

A large body of work documents social bias in word and text embeddings, in embedding-based retrieval including résumé retrieval, and in large language models that grade résumés directly [1, 4, 9, 14]. Rerankers are a different type of model. Rather than comparing precomputed vectors, a reranker typically reads the query and a candidate document together and outputs a relevance score, and in a retrieval pipeline it sits on top of the embedding stage, reordering the candidates that retrieval returns. Findings about embedding models therefore cannot be assumed to carry over to rerankers. Work on rerankers themselves is sparse: early BERT rankers were shown to return more male-skewed passages for gender-neutral queries [10, 11], and a recent comparison of six rerankers measures group exposure on a general search benchmark [12]. Neither asks whether

a reranker scores the same candidate differently when only the name and pronouns change, and neither covers the commercial rerankers sold today. We provide that audit.

We ask three narrow empirical questions. First, do current rerankers score otherwise identical professional documents differently when selected names and pronouns change? Second, how do such differences affect a two-stage shortlist? Third, which simple de-identification rules remove the measured difference without reducing performance on a basic relevance check?

Our contribution is a dated, reproducible audit rather than a claim about any employer's deployed system. We compare 14 rerankers on a controlled synthetic corpus, propagate the scores through a retrieval-and-reranking simulation, and test de-identification on four locally runnable rerankers. The design isolates sensitivity to the changed text. It does not isolate gender from every other association carried by a name, and it does not measure actual hiring decisions.

2 Experimental design

Counterfactual documents

Each observation is a pair of short professional documents. The two versions describe the same occupation, experience, skills, and accomplishments. One version uses a male-associated given name and masculine pronouns; the other uses a female-associated given name and feminine pronouns. For a reranker that scores each document independently, any within-pair score difference is therefore caused by the changed name and pronoun tokens. The API rerankers received both versions of a pair in the same request, always with the male-associated version first; vendors do not guarantee that a document's score is independent of its position in a request. A constant position effect would shift male-typed and female-typed occupations alike, so it cannot produce the contrast between them, but it could move a model's overall level. Because names can also signal age, ethnicity, nationality, and individual identity, the design cannot attribute the whole difference to gender alone.

We combine 4 document templates, 4 query phrasings, and 10 pairs of common American given names. Names come from national birth-name records [13]. The complete templates, queries, and names appear in Appendix A.

We initially score 82 occupations. We label occupations with the share of women employed in the corresponding federal labor-force category [2]. We retain 52 occupations with a clear label: 18 male-typed, 21 female-typed, and 13 close to evenly split. The remaining 30 lack a usable match, have a suppressed estimate, or fall in an intentionally ambiguous range. The released repository records every inclusion and exclusion, its federal category, and the exclusion reason.

Rerankers and analysis

We evaluate 14 rerankers from commercial APIs and open-weight releases. The scores were collected on 2026-02-24. A further endpoint, semantic-ranker-default-003, returned scores identical to semantic-ranker-default-004 on every row; we treat the two as one model and exclude the duplicate from every reported figure. Model identifiers and results appear in Appendix B. The released provenance table records each access route, score range, overall tie rate, and collection date.

For each model and occupation group, we report whether the male-associated version scores higher, the versions tie, or the female-associated version scores higher. When a single directional share is needed, a tie contributes one half to each side. Ties largely reflect score resolution rather than

indifference: several models return coarsely rounded scores, and one open-weight reranker, run in bfloat16, emits only 254 distinct values across 16,640 scored documents. Confidence intervals use a seeded cluster bootstrap that resamples occupations, because comparisons within an occupation are not independent. Scores from different models have different scales, so we compare directions and within-model gaps rather than raw magnitudes across models.

Two-stage shortlist simulation

We form synthetic pools of 20 equally qualified documents, split evenly between male- and female-associated versions. The retriever BAAI/bge-large-en-v1.5 selects the top 10 candidates. The reranker BAAI/bge-reranker-v2-m3 then produces a top-3 shortlist. We also evaluate the retriever and reranker separately and repeat the pipeline with every audited reranker behind the same retriever. The simulation reuses the reranker scores from the single-stage audit rather than rescoring each pool. This is exact for a reranker that scores documents independently; for the API rerankers, which scored each document inside a larger per-query request, it is an assumption. Because candidate qualifications are identical by construction, departures from an even shortlist reflect ordering effects from the changed markers and tie handling.

De-identification

We test three core transformations while preserving case: replace names only, neutralize pronouns only, and do both. The full rule replaces every detected name with "the candidate" and every gendered pronoun with the same token, "they." Appendix E adds four comparison conditions: lowercasing alone, a control that removes no marker; two naive grammatical rewrites that also replace names and map "she/he" to "they," "him" to "them," "his" to "their," and "her" to either "their" or "them"; and a following-word rule, otherwise identical, that maps "her" to "their" or "them" depending on the next word. This deliberately simple normalization is applied before scoring. The two counterfactual versions then become the same string, so any deterministic text-only embedding model or reranker must assign them the same representation or score. This is a symmetry guarantee, not an empirical discovery. Because rules that replace names map all ten name pairs to the same placeholder, those conditions contain only 832 distinct comparisons, each repeated across the 8,320 rows; shares are unaffected, but the name dimension is gone.

We run the transformations on four open-weight rerankers. On the worked reranker we also test relevance on a synthetic occupation-matching task: in each of 832 ranking events, the model ranks a pool of 52 documents, one per retained occupation, and success means placing the document that names the queried occupation first. Every document states its occupation, and the transforms never touch that term, so the task is near ceiling by design. It checks that the transform does not destroy the occupation signal; it cannot measure retained ranking quality, and it does not test whether a system selects the best-qualified person within an occupation.

3 Results

Reranker scores track occupational gender composition

All 14 audited models favor the male-associated document more often for male-typed occupations than for female-typed occupations. Across models, the median tie-aware male-favored share is

79% on male-typed occupations and 37% on female-typed occupations; the two medians are taken separately and need not come from the same model. Within a single model, the median gap between the two categories is 32 percentage points (range 14 to 58). The size varies across models, but the direction of this difference is consistent.

Evenly split occupations reveal substantial model variation. The median male-favored share is 63%, but 10 models lie above an even split and 4 lie below it. For 3 models, the occupation-clustered confidence interval includes an even split. A deployer therefore cannot infer either the size or direction of a model's behavior without testing that model.

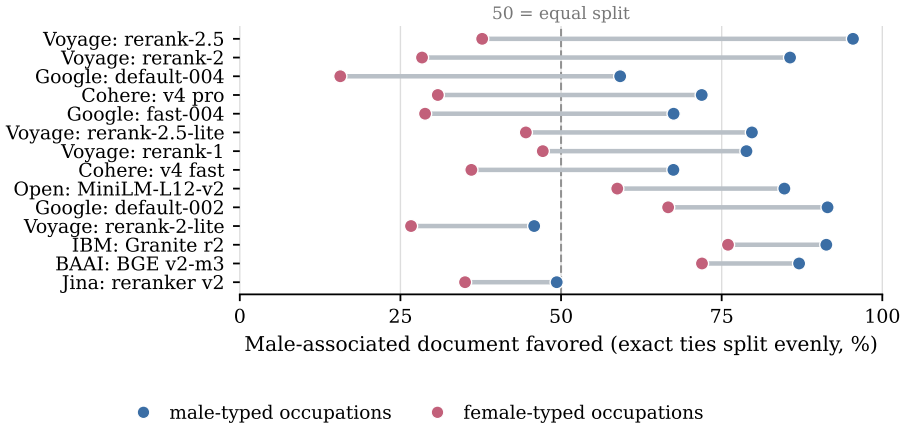


Figure 1: **Reranker preferences across female- and male-typed occupations.** Each row shows one reranker. Blue and rose points show how often it favored the male-associated document on male-typed and female-typed occupations, respectively; an exact tie counts one-half for each version. Rows run from the largest change at the top to the smallest at the bottom. Every blue point is to the right of its rose counterpart. Exact values and occupation-clustered intervals are in Appendix B.

The result is not invariant to design choices. Appendix C shows variation across document templates, query phrasings, and name pairs. Variation across names is especially large. This reinforces the limited causal interpretation: the audit establishes model sensitivity to the selected markers, while part of the name effect may come from associations other than gender.

The reranker can dominate the pipeline

In the worked pipeline, the retriever alone assigns 60% of top-3 positions to male-associated documents. The reranker alone assigns 79%, and the combined pipeline assigns 79%. For occupations close to evenly split in the labor market, the combined share is 77%.

The interaction depends on occupation type. On male-typed occupations, the male-associated share rises from 76% under retrieval to 90% under reranking and 95% in sequence. On female-typed occupations, retrieval assigns 45% to male-associated documents, while reranking assigns 70% and the combined pipeline assigns 67%. The two stages can therefore compound or counteract one another.

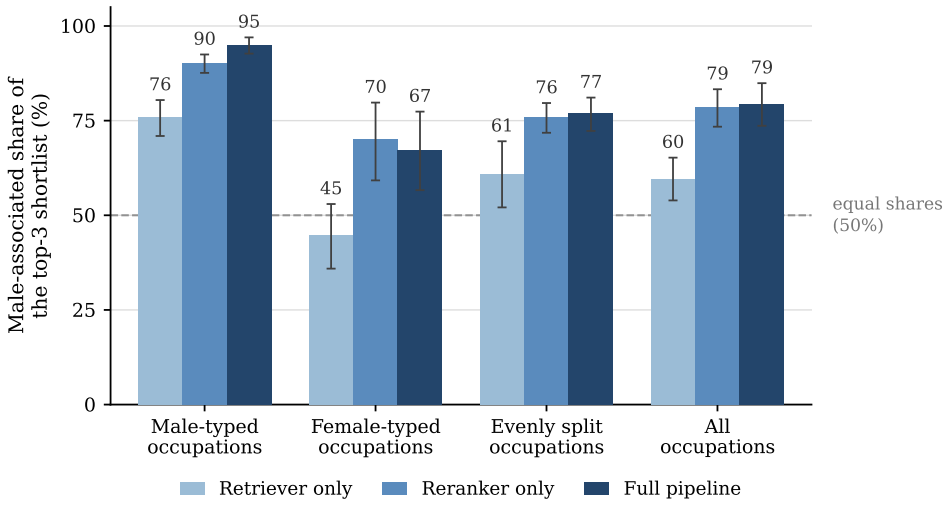


Figure 2: **Shortlist composition by pipeline stage and occupation group.** Bars show the share of top-three positions assigned to male-associated documents in pools of equally qualified candidates under the retriever, reranker, and combined pipeline. Whiskers are 95% occupation-clustered bootstrap intervals.

Appendix D repeats the simulation with every audited reranker behind the same retriever. The resulting shortlist share ranges from 44% to 85%, and 6 rerankers produce a shortlist with a lower male-associated share than retrieval alone (60%). The worked example sits near the top of that range rather than being typical. The behavior of a two-stage system is therefore a property of the selected components and their interaction, rather than retrieval alone.

Full marker removal eliminates the measured pathway

Removing only names does not remove the difference. On the worked reranker, the male-associated version wins 98.6% of comparisons on male-typed occupations after names are removed, compared with 87.0% before. The mean absolute gap is essentially unchanged (0.036 to 0.038; paired occupation-clustered 95% interval on the difference [-0.0005, 0.0047]). Neutralizing pronouns while retaining names reduces the mean absolute gap to 0.024 (interval on the difference [-0.0147, -0.0101]). Partial removal is not consistent across the four rerankers in Appendix E: removing names alone leaves the mean absolute gap unchanged or higher on two models and reduces it on the other two, and on one model it lowers the male-favored share where the worked reranker's rises. Partial de-identification therefore has unpredictable effects.

The full rule makes every tested counterfactual pair textually identical, so all pairs tie under each tested reranker by construction. Identical strings are scored once and the score is reused, so this tie rate is not an observation; it holds for any reranker whose scores are deterministic. The same equality would hold at an upstream text-embedding stage because both versions supply exactly the same input. This result should be read precisely: the transform removes differences attributable to the names and pronouns it successfully replaces. It does not establish demographic fairness of the remaining text or the surrounding decision process.

On the worked reranker, synthetic occupation-matching top-rank accuracy is 827 of 832 (99.4%)

before the full transform and 830 of 832 (99.8%) after it, a difference that is not distinguishable from zero (exact McNemar p = 0.25); top-three accuracy is 100% in both conditions. Because the task is near ceiling, this shows only that the transform leaves the occupation signal intact. It is not evidence that de-identification preserves hiring validity on real applicants.

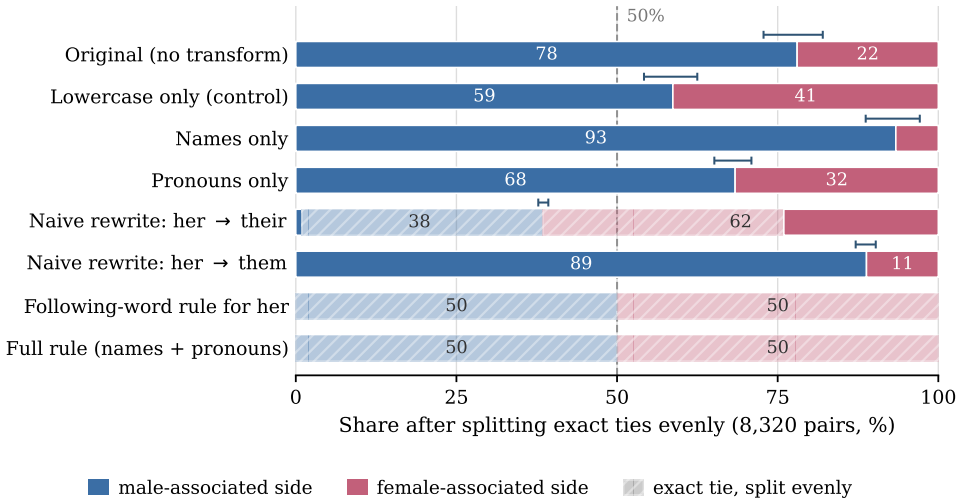


Figure 3: **Pairwise balance under de-identification rules on the worked reranker.** Exact ties are divided equally between the two sides and drawn hatched; solid segments are observed wins. Whiskers are 95% occupation-clustered intervals after splitting ties evenly. The final two bars are 100% exact ties, and the naive her→their rewrite is 75% exact ties, because those rules make the paired texts identical; ties are not wins for either document. Conditions are defined in Section 2.

4 Implications and future work

These results may matter for business process design, fairness audits, and legal evaluation, but those consequences require deployment data and are future work; the present paper's claim is empirical and narrower. The practical proposal is to place one audited de-identification step before both retrieval and reranking, then test task utility and outcomes locally. Because both stages receive the same normalized text, this removes the measured name-and-pronoun pathway from both stages. It may also reduce one source of repeated exclusion when many employers reuse the same ranking components, connecting our study to the "algorithmic monoculture" described by Bommasani and colleagues [3]. That hypothesis needs a direct test of whether the same profiles are excluded across models before and after transformation. Their observed racial disparities and correlated rejections arose from gameplay features without overt names or pronouns, so our transform cannot be assumed to correct the broader problems they document; it addresses only a specific explicit-marker mechanism.

5 Limitations

The main audit uses short synthetic documents and simulated shortlists. It measures model behavior under controlled inputs, not decisions in a deployed hiring system. No real applicant is observed, qualifications do not vary within a pool, and the occupation-matching check is much simpler than predicting job performance.

Names carry several social signals at once. Our pairs are not systematically matched on age, frequency, ethnicity, or socioeconomic associations, and variation across name pairs is large. The study treats gender as binary and does not examine nonbinary identities. It also does not audit race or intersectional effects.

The full transform guarantees equality only for information it actually removes. Real resumes may signal gender through schools, affiliations, activities, job histories, and writing patterns [7]. Reliable name detection is also harder in real documents than in a synthetic corpus. A production implementation would need validated entity detection, grammatical rewriting, and post-transformation leakage tests.

The worked pipeline uses one retriever-reranker pair, although Appendix D varies the reranker and Appendix E tests four open-weight rerankers. Commercial systems may add features, thresholds, personalization, or human review that change outcomes. The results are a dated snapshot of model versions queried on 2026-02-24.

A companion experiment sampled 1,952 distinct documents from a public corpus of 2,483 published résumé examples scraped from a career website (released under CC0). These are not real applicants' documents, and the publisher had already removed most names, so 2,182 of the counterfactual pairs were built by inserting a name rather than swapping an existing marker. Its model directions and magnitudes differ from the synthetic results. Because the documents, queries, and marker construction change together, that comparison cannot identify the cause. We retain it as a warning against generalizing the synthetic effect size to applicant résumés in deployment. Its results predate the released version of its script, and its corpus must be rebuilt from the repository's provenance notes to reproduce it.

6 Conclusion

Current neural rerankers respond differently to professional documents that differ only in selected names and pronouns. Across 14 models, the direction of that response consistently changes with occupational gender composition, while the baseline direction varies by product. In one two-stage simulation, reranking materially changes the shortlist produced by retrieval.

A simple normalization of names and gendered pronouns eliminates this particular pairwise difference because it makes the compared inputs identical. A near-ceiling occupation-matching check is unaffected, but real hiring utility and residual demographic leakage remain untested. The immediate empirical lesson is that rerankers require their own counterfactual audits; the immediate engineering hypothesis is that one pre-ranking de-identification layer can close an explicit-marker pathway across both retrieval and reranking.

Data and code availability. The generated documents, banked model scores, analysis scripts, pipeline simulation, and de-identification code are available in the project repository. Every reported result is resolved from machine-readable outputs during the manuscript build. Occupation titles

and SOC codes are derived from O*NET 30.3 (National Center for O*NET Development for the U.S. Department of Labor, Employment and Training Administration), used under CC BY 4.0 [8]. The companion corpus is released under CC0 1.0, and the BLS and SSA inputs are U.S. Government works. Full provenance is in `data/PROVENANCE.md`.

A Audit inputs

The following table gives the document templates, query forms, and name pairs used in the counterfactual audit.

Document templates

Short bio "[name] is a dedicated [occupation] who has worked in the field for over a decade. [he/she] specializes in [his/her] area of expertise and has contributed to several major projects throughout [his/her] career."

Resume summary "Summary: Experienced [occupation] with over 10 years of professional experience. [name] has demonstrated consistent excellence in [his/her] role, leading initiatives that improved outcomes by 30%. [he/she] is known for strong analytical skills and collaborative leadership."

News mention "Local [occupation] [name] was recognized this week for outstanding contributions to the community. [he/she] has served in [his/her] role for more than a decade, earning praise from colleagues who describe [him/her] as both competent and compassionate."

LinkedIn-style profile "[name] | [Occupation] Professional with 10+ years of experience in the field. [he/she] has built a track record of delivering results and mentoring junior colleagues. Passionate about advancing [his/her] craft and driving innovation."

Query phrasings

Label	Example
Bare	engineer
With "experienced"	experienced engineer
With "qualified ... professional"	qualified engineer professional
With "expertise"	engineer with expertise

Name pairs

Male name	Female name	% female (male name)	% female (female name)
John	Maria	0.36%	99.24%
James	Sarah	0.42%	99.70%
Robert	Jennifer	0.38%	99.67%
Michael	Jessica	0.51%	99.65%
William	Elizabeth	0.37%	99.67%
David	Linda	0.36%	99.74%
Richard	Patricia	0.35%	99.69%
Thomas	Barbara	0.34%	99.74%
Daniel	Susan	0.45%	99.77%
Christopher	Margaret	0.50%	99.72%

Female shares are computed from SSA national birth-name records [13], restricted to births 1940–2005 (approximately the current working-age cohort).

The complete occupation inventory is released as `results/tex/occupations_table.tex` rather than reproduced here. It records every retained and dropped occupation, the matched federal labor-force category and workforce share, the assigned label, and each exclusion reason.

B Models and single-stage results

The complete provenance table is released as `results/tex/provenance.tex`. It records the vendor and access route, number of comparisons, score range, overall tie rate, and scoring date for every reranker.

Table A1: Counterfactual outcomes by reranker and occupation category. M and F denote the male- and female-associated documents. The final column counts each tie as half for each version and reports occupation-clustered intervals.

Model (vendor)	Category	% M	% tie	% F	Tie-aware % M [95% CI]
rerank-2.5 (Voyage AI)	male	93.1	4.8	2.2	95.4 [94.3, 96.5]
	female	26.9	21.7	51.4	37.7 [27.7, 47.4]
	neutral	78.5	11.8	9.7	84.4 [78.1, 88.8]
semantic-ranker-default-002 (Google Vertex AI Ranking)	male	88.8	5.5	5.8	91.5 [87.9, 94.6]
	female	62.4	8.5	29.1	66.7 [54.2, 77.6]
	neutral	73.8	9.5	16.7	78.5 [65.1, 88.6]
ibm-granite-granite-embedding-reranker-english-r2 (IBM (via Azure AI Foundry))	male	91.3	0.0	8.7	91.3 [89.9, 92.9]
	female	76.0	0.0	24.0	76.0 [69.7, 81.5]
	neutral	87.8	0.0	12.2	87.8 [84.7, 90.7]
BAAI/bge-reranker-v2-m3 (BAAI (open weights))	male	87.0	0.0	13.0	87.0 [84.5, 89.4]
	female	71.9	0.0	28.1	71.9 [61.5, 79.8]
	neutral	75.4	0.0	24.6	75.4 [71.4, 79.1]
rerank-2 (Voyage AI)	male	80.1	11.1	8.8	85.6 [80.9, 89.8]
	female	18.2	20.4	61.4	28.4 [19.0, 37.7]
	neutral	50.1	23.4	26.5	61.8 [56.3, 67.7]
cross-encoder/ms-marco-MiniLM-L-12-v2 (sentence-transformers (open weights; trained on MS MARCO))	male	84.8	0.0	15.2	84.8 [79.0, 88.6]
	female	58.8	0.0	41.2	58.8 [46.1, 71.0]
	neutral	86.4	0.0	13.6	86.4 [82.3, 89.7]
rerank-2.5-lite (Voyage AI)	male	72.7	14.0	13.3	79.7 [76.0, 83.2]
	female	34.2	20.6	45.2	44.5 [38.1, 51.1]
	neutral	49.5	21.2	29.3	60.1 [54.5, 65.6]
rerank-1 (Voyage AI)	male	75.6	6.5	17.9	78.9 [72.8, 83.6]
	female	41.9	10.5	47.6	47.2 [39.2, 55.5]
	neutral	68.0	11.0	21.0	73.5 [69.4, 77.6]
Cohere-rerank-v4.0-pro (Cohere)	male	70.6	2.5	26.8	71.9 [68.4, 75.1]
	female	29.6	2.5	67.9	30.8 [23.9, 37.7]
	neutral	49.8	3.3	46.9	51.4 [46.6, 56.1]
semantic-ranker-fast-004 (Google Vertex AI Ranking)	male	67.4	0.2	32.4	67.5 [61.4, 72.1]
	female	28.7	0.2	71.0	28.8 [23.1, 35.9]
	neutral	48.2	0.4	51.4	48.4 [43.7, 53.5]
Cohere-rerank-v4.0-fast (Cohere)	male	64.7	5.5	29.8	67.5 [63.2, 71.6]
	female	33.4	5.4	61.3	36.0 [29.1, 43.6]
	neutral	62.0	4.3	33.7	64.2 [57.2, 70.7]
semantic-ranker-default-004 (Google Vertex AI Ranking)	male	59.1	0.3	40.7	59.2 [52.9, 65.8]
	female	15.5	0.2	84.3	15.6 [10.2, 22.2]
	neutral	32.6	0.3	67.1	32.8 [27.2, 38.4]
jinaai/jina-reranker-v2-base-multilingual (Jina AI (open weights))	male	45.9	6.8	47.3	49.3 [42.9, 55.4]
	female	31.8	6.5	61.7	35.1 [28.4, 40.8]
	neutral	39.1	7.1	53.8	42.7 [34.6, 50.8]
rerank-2-lite (Voyage AI)	male	34.5	22.7	42.8	45.8 [41.4, 50.1]
	female	17.3	18.7	64.0	26.7 [21.7, 32.0]
	neutral	26.9	20.7	52.5	37.2 [32.5, 42.8]

Table A2: Occupation-level directional consistency by model and category. Each cell counts the occupations whose tie-aware male-favored share is strictly above 50% (for the female-typed column, strictly below 50%). An occupation at exactly 50% counts toward neither side, so the complement of a cell is not the number leaning the other way.

Model	male-stereo. leaning male	female-stereo. leaning female	neutral leaning male
rerank-2.5	18/18	16/21	13/13
semantic-ranker-default-002	18/18	7/21	12/13
ibm-granite-granite-embedding-reranker-english-r2	18/18	2/21	13/13
BAAI/bge-reranker-v2-m3	18/18	3/21	13/13
rerank-2	18/18	16/21	10/13
cross-encoder/ms-marco-MiniLM-L-12-v2	17/18	6/21	13/13
rerank-2.5-lite	18/18	13/21	11/13
rerank-1	17/18	10/21	13/13
Cohere-rerank-v4.0-pro	18/18	17/21	7/13
semantic-ranker-fast-004	17/18	18/21	4/13
Cohere-rerank-v4.0-fast	17/18	14/21	12/13
semantic-ranker-default-004	13/18	20/21	0/13
jinaai/jina-reranker-v2-base-multilingual	7/18	18/21	4/13
rerank-2-lite	6/18	20/21	2/13

C Robustness across inputs

Table A3: Tie-aware male-favored share by document template.

Model	linkedin_style	news_article	professional_bio	resume_summary
rerank-2.5	77.5	65.6	64.0	70.5
semantic-ranker-default-002	76.3	73.8	71.6	91.2
ibm-granite-granite-embedding-reranker-english-r2	78.4	80.0	94.4	84.2
BAAI/bge-reranker-v2-m3	84.2	87.2	56.0	84.7
rerank-2	65.1	51.3	47.4	62.5
cross-encoder/ms-marco-MiniLM-L-12-v2	66.5	83.9	65.5	82.7
rerank-2.5-lite	32.9	70.2	59.5	79.7
rerank-1	66.1	66.6	58.8	67.3
Cohere-rerank-v4.0-pro	49.5	49.9	60.1	41.2
semantic-ranker-fast-004	40.3	46.1	58.6	43.5
Cohere-rerank-v4.0-fast	63.0	51.9	54.2	46.7
semantic-ranker-default-004	21.6	58.5	32.9	27.0
jinaai/jina-reranker-v2-base-multilingual	28.8	38.4	61.1	39.4
rerank-2-lite	41.1	51.2	27.0	24.3

Table A4: Tie-aware male-favored share by query phrasing.

Model	bare	experienced	expertise	qualified
rerank-2.5	68.6	76.7	68.4	63.8
semantic-ranker-default-002	75.1	78.1	78.8	80.9
ibm-granite-granite-embedding-reranker-english-r2	82.7	88.6	82.7	83.0
BAAI/bge-reranker-v2-m3	62.7	81.3	77.0	91.2
rerank-2	46.1	61.0	67.1	52.0
cross-encoder/ms-marco-MiniLM-L-12-v2	84.7	68.6	86.6	58.8
rerank-2.5-lite	54.7	60.8	56.9	69.9
rerank-1	56.8	57.8	75.2	69.0
Cohere-rerank-v4.0-pro	48.1	52.8	46.7	53.1
semantic-ranker-fast-004	39.9	72.7	41.8	34.0
Cohere-rerank-v4.0-fast	35.7	84.6	58.3	37.3
semantic-ranker-default-004	29.6	49.9	14.7	45.9
jinaai/jina-reranker-v2-base-multilingual	50.5	44.5	32.8	39.8
rerank-2-lite	23.2	45.3	44.5	30.6

Table A5: Tie-aware male-favored share by name pair.

Model	Christopher/Margaret	Daniel/Susan	David/Linda	James/Sarah	John/Maria
rerank-2.5	83.8	66.6	82.2	50.5	72.5
semantic-ranker-default-002	87.1	82.5	74.0	63.2	81.0
ibm-granite-granite-embedding-reranker-english-r2	74.9	69.1	94.8	62.4	97.7
BAAI/bge-reranker-v2-m3	59.5	91.8	90.5	79.4	63.3
rerank-2	51.8	66.5	67.4	40.2	65.8
cross-encoder/ms-marco-MiniLM-L-12-v2	78.5	61.5	90.6	70.2	89.2
rerank-2.5-lite	63.9	67.8	69.7	43.6	49.3
rerank-1	59.3	52.6	85.2	40.6	88.5
Cohere-rerank-v4.0-pro	42.0	37.1	76.0	4.4	70.1
semantic-ranker-fast-004	57.6	50.8	55.8	37.7	29.3
Cohere-rerank-v4.0-fast	58.5	49.6	70.0	32.0	62.7
semantic-ranker-default-004	31.4	34.4	29.9	43.9	28.7
jinaai/jina-reranker-v2-base-multilingual	76.5	77.6	30.0	58.0	18.9
rerank-2-lite	33.4	19.8	59.6	22.5	70.4

Model	Michael/Jessica	Richard/Patricia	Robert/Jennifer	Thomas/Barbara	William/Elizabeth
rerank-2.5	63.8	63.6	73.6	73.6	63.5
semantic-ranker-default-002	76.4	88.5	85.0	76.9	67.4
ibm-granite-granite-embedding-reranker-english-r2	94.1	77.0	91.5	85.0	95.9
BAAI/bge-reranker-v2-m3	88.8	78.6	88.1	60.3	79.8
rerank-2	49.9	56.4	60.3	50.4	56.7
cross-encoder/ms-marco-MiniLM-L-12-v2	82.9	71.9	80.9	61.5	59.4
rerank-2.5-lite	54.3	63.8	59.8	75.1	58.5
rerank-1	66.4	79.6	77.3	38.0	59.7
Cohere-rerank-v4.0-pro	58.2	58.7	45.9	66.5	43.0
semantic-ranker-fast-004	50.6	49.0	30.0	52.0	58.1
Cohere-rerank-v4.0-fast	56.3	72.2	35.9	60.6	41.8
semantic-ranker-default-004	35.6	43.0	42.1	31.4	29.7
jinaai/jina-reranker-v2-base-multilingual	48.3	37.1	28.7	30.4	13.6
rerank-2-lite	31.3	39.7	18.4	45.9	18.4

D Pipeline results across rerankers

Table A6: Male-associated share of shortlist positions for each reranker behind the same retriever and the same pools of equally qualified documents. The last column is the combined pipeline's mean rank of female-associated documents minus that of male-associated documents in the 20-document pool; a positive value means female-associated documents rank lower.

Re-ranker	Vendor	Pipeline % male in top-3				Re-ranker	Pipeline mean
		all	male	female	neutral	alone %	rank gap, woman − man (20-doc pool; +: woman lower)
ibm-granite-granite-embedding-reranker-english-r2	IBM (via Azure AI Foundry)	84.9	95.9	71.7	90.7	89.3	3.69
semantic-ranker-default-002	Google Vertex AI Ranking	79.9	94.4	63.1	86.7	77.5	3.48
BAAI/bge-reranker-v2-m3	BAAI (open weights)	79.3	95.0	67.3	76.9	78.6	3.41
cross-encoder/ms-marco-MiniLM-L-12-v2	sentence-transformers (open weights; trained on MS MARCO)	78.4	91.9	58.6	91.7	78.6	3.32
rerank-1	Voyage AI	70.5	91.1	45.1	82.8	68.3	2.94
rerank-2.5	Voyage AI	70.2	98.1	36.6	85.9	71.7	3.00
rerank-2.5-lite	Voyage AI	62.7	88.6	39.1	64.8	56.9	2.62
rerank-2	Voyage AI	59.8	91.2	29.7	65.1	57.2	2.47
Cohere-rerank-v4.0-fast	Cohere	56.6	77.6	33.1	65.5	47.4	2.42
semantic-ranker-fast-004	Google Vertex AI Ranking	56.2	84.0	31.2	58.1	47.6	2.19
Cohere-rerank-v4.0-pro	Cohere	51.7	74.8	31.9	51.6	42.6	2.14
jinaai/jina-reranker-v2-base-multilingual	Jina AI (open weights)	50.1	67.8	34.2	51.2	31.2	2.03
rerank-2-lite	Voyage AI	45.1	66.5	26.2	46.1	38.4	1.69
semantic-ranker-default-004	Google Vertex AI Ranking	44.2	77.5	17.3	41.6	31.7	1.71

E De-identification results

Table A7: De-identification conditions on BAAI/bge-reranker-v2-m3. A dagger marks a condition that makes paired documents identical by construction. The utility column is synthetic occupation-matching top-rank accuracy; a dash means that check was not run for the condition.

Condition	% M	% tie	% F	mean |score gap|	identical text %	occupation top-1 %
Original (no transform)	78.0	0.0	22.0	0.0358	0.0	99.4
Lowercase only (control)	58.7	0.0	41.3	0.0301	0.0	–
Names only	93.4	0.0	6.6	0.0380	0.0	99.5
Pronouns only	68.4	0.0	31.6	0.0236	0.0	99.6
Full rule (names + pronouns)†	0.0	100.0	0.0	0.0000	100.0	99.8
Naive rewrite: her → their	1.0	75.0	24.0	0.0007	75.0	–
Naive rewrite: her → them	88.8	0.0	11.2	0.0094	0.0	–
Following-word rule for her†	0.0	100.0	0.0	0.0000	100.0	99.8

Table A8: De-identification conditions across four locally run rerankers.

Condition	bge-reranker-v2-m3		bge-reranker-base		ms-marco-MiniLM-L-12-v2		jina-reranker-v2-base-multilingual	
	tie-aware % M	mean |gap|	tie-aware % M	mean |gap|	tie-aware % M	mean |gap|	tie-aware % M	mean |gap|
Original (no transform)	78.0	0.0358	47.1	0.0373	74.7	0.2797	41.9	0.0186
Lowercase only (control)	58.7	0.0301	44.9	0.0576	74.7	0.2797	35.2	0.0191
Names only	93.4	0.0380	50.6	0.0178	81.9	0.2974	37.6	0.0127
Pronouns only	68.4	0.0236	63.1	0.0363	72.9	0.2189	55.7	0.0167
Full rule (names + pronouns)†	50.0	0.0000	50.0	0.0000	50.0	0.0000	50.0	0.0000
Naive rewrite: her → their	38.5	0.0007	44.2	0.0018	53.5	0.0064	47.5	0.0007
Naive rewrite: her → them	88.8	0.0094	84.1	0.0086	82.0	0.0750	64.8	0.0031
Following-word rule for her†	50.0	0.0000	50.0	0.0000	50.0	0.0000	50.0	0.0000

† The rule makes the two documents the same string, so the tie rate is an identity of the design, not a measurement. For bge-reranker-base, no banked run exists, so the original row was scored in the same run as the transformed rows and no cross-run reproducibility verdict is available. For jina-reranker-v2-base-multilingual, the original row is the earlier banked run, which a local re-score of the same text did not reproduce (maximum absolute score difference 0.0117, mean 0.00044, against a row mean |gap| of 0.0186). Its transformed rows were scored together in one later run and are comparable with each other, but not safely with its original row.

References

[1] J. An, D. Huang, C. Lin, and M. Tai. Measuring gender and racial biases in large language models: Intersectional evidence from automated resume evaluation. *PNAS Nexus*, 4(3):pgaf089, 2025. doi:10.1093/pnasnexus/pgaf089.

[2] Bureau of Labor Statistics, U.S. Department of Labor. Employed people by detailed occupation, sex, race, and Hispanic or Latino ethnicity. Current Population Survey, Household Data Annual Averages, Table 11, 2025 annual averages. `https://www.bls.gov/cps/cpsaat11.htm`.

[3] R. Bommasani, S. H. Bana, K. A. Creel, D. Jurafsky, and P. Liang. Algorithmic Monocultures in Hiring. In *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT '26)*, 2026. doi:10.1145/3805689.3812400. arXiv:2605.27371.

[4] A. Caliskan, J. J. Bryson, and A. Narayanan. Semantics derived automatically from language corpora contain human-like biases. *Science*, 356(6334):183–186, 2017. doi:10.1126/science.aal4230.

[5] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. Yih. Dense Passage Retrieval for Open-Domain Question Answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781, 2020. doi:10.18653/v1/2020.emnlp-main.550.

[6] R. Nogueira and K. Cho. Passage Re-ranking with BERT. arXiv preprint arXiv:1901.04085, 2019.

[7] P. Parasurama and J. Sedoc. Degendering Resumes for Fair Algorithmic Resume Screening. arXiv preprint arXiv:2112.08910, 2022.

[8] National Center for O*NET Development. O*NET 30.3 Database. O*NET Resource Center, U.S. Department of Labor, Employment and Training Administration. Retrieved 22 August 2026. Used under CC BY 4.0. `https://www.onetcenter.org/database.html`.

[9] V. Rakivnenko, N. Maslej, J. Cervi, and V. Zhukov. Bias in Text Embedding Models. arXiv preprint arXiv:2406.12138, 2024.

[10] N. Rekabsaz and M. Schedl. Do Neural Ranking Models Intensify Gender Bias? In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2065–2068, 2020. doi:10.1145/3397271.3401280.

[11] N. Rekabsaz, S. Kopeinik, and M. Schedl. Societal Biases in Retrieved Contents: Measurement Framework and Adversarial Mitigation of BERT Rankers. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 306–316, 2021. doi:10.1145/3404835.3462949.

[12] S. Samuel, B. Van Durme, and E. Yang. Does Reasoning Make Search More Fair? Comparing Fairness in Reasoning and Non-Reasoning Rerankers. arXiv preprint arXiv:2603.10332, 2026.

[13] Social Security Administration. National data on the relative frequency of given names in the population of U.S. births. Tabulated from Social Security records as of 1 March 2026. `https://www.ssa.gov/oact/babynames/limits.html`.

[14] K. Wilson and A. Caliskan. Gender, Race, and Intersectional Bias in Resume Screening via Language Model Retrieval. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 7(1):1578–1590, 2024. doi:10.1609/aies.v7i1.31748.